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Fast model predictive control of power amplifiers for nanometer precision motion systems

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***Abstract*— This paper deals with the design of a fast MPC algorithm for the current control of a power amplifier utilized for nanometer precision positioning systems within lithography machines. In order to achieve nanometer precision positioning, the internal power amplifier must accurately track a current reference in a very short time (tens of microseconds). Classical industrial control solutions based on transfer functions do not take duty–cycle limits into account and suffer from limited bandwidth, which in turn limits the achievable positioning precision. We design a fast gradient based MPC algorithm that can accurately track the dynamic current reference while satisfying constraints. Simulations show that the MPC-controlled amplifier results in at least 2x better nanometer positioning precision for specific metrics employed in the lithography industry, compared to an industrial loop-shaping controller and an LQR controller.**

***Index Terms*— Constrained control, Model predictive control, Fast gradient methods, Power amplifiers, Motion control**

I. Introduction

In the semiconductor lithography industry, Moore's law dictates that the number of transistors in a dense integrated circuit doubles about every two years [1]. Thus, future generations of industrial positioning systems are facing an increasing demand in precision and throughput [2]. In order to achieve the 2 nm wafer stage precision, the position error introduced by the motion system is required to hold only a small (e.g. $\pm 1\%$) fraction of the total position error [3], [4].

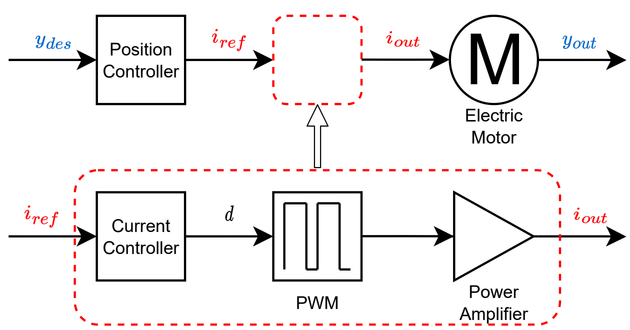

Fig. 1: Generic outline of a motion control system.

Fig. 1 depicts the generic outline of a high-precision motion control system, which consists of two parts: an outer position control loop and an inner current control loop. For the outer position control loop, the position controller calculates an appropriate current reference signal to actuate the electric motor based on the desired position profile. Then, for the inner current control loop, the switched power amplifier is controlled to generate the corresponding current signal. Conventionally, since the current loop is sampled and operated much faster compared to the position loop, the impact of the current loop on the position tracking accuracy is neglected. However, for high-precision applications, any imperfection in the current loop performance adversely affects the overall position system accuracy [5].

For power amplifier control, classical PWM control design methods for power electronics based on an averaged model (duty-cycle) are employed [6]. The main limitation of this approach comes from ignoring the hard duty-cycle constraints, which results in limited bandwidth and tracking performance due to saturation. A cascaded control structure consisting of an inner observer-based LQR controller and an outer input-output SISO controller was proposed in [7] to increase the bandwidth. This results in a faster response, but still suffers from saturation, which can lead to undesirable current overshoots.

In this paper, we aim to solve both the bandwidth and saturation issues by using model predictive control (MPC) for output current control. Normally, the application of MPC in power electronics is limited by the computation time, which is very short, e.g., 2.5 microseconds for the considered power amplifier. However, recent contributions involving both efficient MPC designs and fast solvers for MPC, indicate that it is possible to meet these timing constraints, see, e.g., [8]–[13]. The adopted control architecture consists of a feedforward controller that inverts the amplifier load, an outer SISO feedback controller that shapes the frequency response of the MPC closed-loop system, and an inner MPC controller with feedforward preview for current control, which ensures that the total duty-cycle sent to the PWM stage is always within constraints. To meet the strict timing requirements we adopt the fast gradient method (FGM) from [8] and we propose an optimized initialization (warm-start) and a more effective stopping criterion for the considered power amplifier. The resulting computational times are around 65 microseconds, but the actual time is expected to be lower, as we are close to the accuracy of the Matlab tic-toc functions. Based on the simple FGM operations and the number of iterations, which never exceeds 5, we can anticipate successful real-time implementation on FPGA, see also [9]. Simulation of the MPC-controlled power amplifier within the position control loop indicates at least 2 times improvement with respect to classical SISO control and LQR control.

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II. Preliminaries

Fig. 2 depicts the topology of the considered high-precision industrial switched power amplifier, which has a symmetrical structure on the positive and negative sides. V_{bus} is the supply voltage. L, C, R are the inductor, capacitor and resistor of the LC filters. The actuator is modelled with a resistive load R_m and an inductive load L_m, where the output current i_o is used to actuate the motor [14]. On each side,

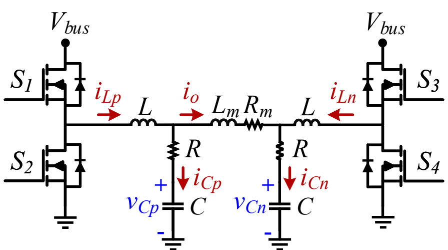

Fig. 2: Topology of the considered power amplifier.

the power amplifier is switched with a separate PWM which has a fixed sampling frequency f_s. The two switches on the same side cannot be turned on and off at the same time. The corresponding duty-cycle inputs on both sides are denoted as $u = [d_p \quad d_n]^\top$ and they are strictly limited within the box constraint $u_{min} \leq u \leq u_{max}$. The averaged continuous-time state-space model (1) can be derived as:

$$\begin{aligned} \dot{x}(t) &= A_c x(t) + B_c u(t), \\ y(t) &= C_c x(t), \end{aligned} \tag{1}$$

where

$$A_c = \begin{bmatrix} -\frac{R}{L} & -\frac{1}{L} & 0 & 0 & \frac{R}{L} \\ \frac{1}{C} & 0 & 0 & 0 & -\frac{1}{C} \\ 0 & 0 & -\frac{R}{L} & -\frac{1}{L} & -\frac{R}{L} \\ 0 & 0 & \frac{1}{C} & 0 & \frac{1}{C} \\ \frac{R}{L_m} & \frac{1}{L_m} & -\frac{R}{L_m} & -\frac{1}{L_m} & -\frac{2R+R_m}{L_m} \end{bmatrix},$$

$$B_c = \begin{bmatrix} \frac{V_{bus}}{L} & 0 \\ 0 & 0 \\ 0 & \frac{V_{bus}}{L} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, C_c = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix}.$$

The system states are $x = [i_{Lp} \quad v_{Cp} \quad i_{Ln} \quad v_{Cn} \quad i_o]^\top$. The measured outputs are $y = [i_{Cp} \quad i_{Cn} \quad i_o]^\top$.

The averaged model (1) is discretized using the zero-order hold (ZOH) method with $T_s = 1/f_s = 2.5\mu\text{s}$, which is

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k), \\ y(k) &= Cx(k). \end{aligned} \tag{2}$$

Next, we recall the cascaded LQR control structure proposed in [7], with an additional feedforward controller for inverting the load, which will serve as a basis for introducing the MPC control structure proposed in this paper.

A. LQR cascaded current control scheme

Fig. 3 presents an LQR cascaded current control scheme for switched power amplifiers, which aims to ensure both good transient and stationary performance.

1) Inner loop: In order to achieve fast reference tracking with a good transient response, the inner closed-loop (dashed line block) employs LQR to stabilize the system at the equilibrium point. At steady-state, the reference states and inputs (x_s, u_s) can be calculated from the reference r_o as

$$x_s = M_x r_o + N_x, \quad u_s = M_u r_o + N_u, \tag{3}$$

where $M_x = \begin{bmatrix} 1 & \frac{R_m}{2} & -1 & -\frac{R_m}{2} & 1 \end{bmatrix}^\top$, $N_x = \begin{bmatrix} 0 & \frac{V_{bus}}{2} & 0 & \frac{V_{bus}}{2} & 0 \end{bmatrix}^\top$, $M_u = \begin{bmatrix} \frac{R_m}{2V_{bus}} & -\frac{R_m}{2V_{bus}} \end{bmatrix}^\top$ and $N_u = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix}^\top$. The LQR controller is designed to minimize the following cost function:

$$\sum_{i=0}^{\infty} (x(i) - x_s)^\top Q (x(i) - x_s) + (u(i) - u_s)^\top R (u(i) - u_s), \tag{4}$$

where Q and R are the weight matrices. A Luenberger observer with gain L_o is used to estimate the full state, i.e.

$$L_o = \begin{bmatrix} 1.34 & 0.13 & 0 \\ -2.68 & -0.15 & 0 \\ -1.34 & 0 & 0.13 \\ 2.68 & 0 & -0.15 \\ 0.98 & 0 & 0 \end{bmatrix}. \tag{5}$$

2) Outer loop: The outer loop aims to diminish the steady-state error and accomplish frequency domain specifications. The inner closed-loop with the LQR controller and the amplifier model is treated as a separate SISO system with r_o as input and i_o as output. Based on the separation principle [15] for linear systems, the observer is not considered in the outer loop design. A second-order feedback controller is designed using the loop shaping method. The discrete transfer function of the feedback controller $C_{FB}(z)$ is

$$C_{FB}(z) = \frac{48.69z^2 - 95.87z + 47.2}{z^2 - 2z + 1}. \tag{6}$$

3) Feedforward controller and input saturator: In addition to the LQR and the feedback controller, a feedforward controller is also included in the control scheme, which helps to achieve a faster and more accurate tracking performance. Due to the symmetrical structure, a common duty cycle d_{com} is introduced as the system input, which satisfies $d_p = 0.5 + 0.5d_{com}$ and $d_n = 0.5 - 0.5d_{com}$. The power amplifier (1) can also be modelled as the following transfer function

$$\frac{i_{out}(s)}{d_{com}(s)} \approx \frac{V_{bus}}{sL_m + R_m} \frac{sRC + 1}{s^2 LC + 1}, \tag{7}$$

The feedforward controller is designed by only taking the inverse of the low frequency load part $\frac{V_{bus}}{sL_m + R_m}$. To avoid non-causality, a roll-off pole is added. The discrete transfer function of the feedforward controller $C_{FF}(z)$ is:

$$C_{FF}(z) = \frac{55.56z - 55.53}{z - 0.082}. \tag{8}$$

As shown in Fig. 3, the sum of the 3 input signals (u_{FF}, u_s, u_{LQR}) is the optimal input for the power amplifier. A saturator is added afterwards to satisfy the physical limit of the duty-cycle signal. Compared to the feedback controller,

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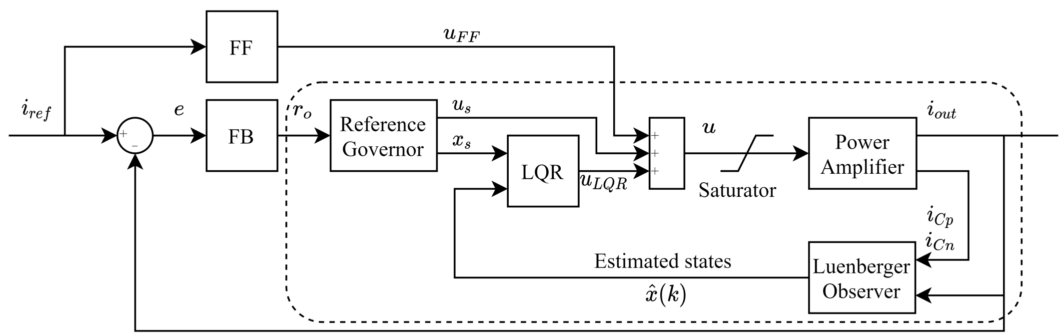

Fig. 3: LQR cascaded current control scheme.

the feedforward controller has a faster and more drastic reaction to a sudden reference signal change. Thus, the sum of the 3 input signals easily reaches saturation limits when the reference signal is updated from the position controller. As it will be shown in the simulation section, saturation has a significant impact on position precision. Therefore, in this paper we will next develop an MPC structure for controlling the output current and guaranteeing that the total input applied to the PWM stage is always within constraints.

III. Main results

The proposed MPC architecture is presented in Fig. 4. Since MPC enables constrained control, the saturator in Fig. 3 is not needed.

A. MPC design for the cascaded current control scheme

To define the MPC optimization problem, let $x_{0|k} = \hat{x}(k)$ denotes the estimated state at time k and let $x_{i|k}$ and $u_{i|k}$ denote the predicted state and input at time $k+i$ over a prediction horizon N. The sequence of the predicted inputs U_k can be expressed as

$$U_k = \left[u_{0|k}^\top, \dots, u_{N-1|k}^\top\right]^\top. \tag{9}$$

Similarly, since the output current reference i_{ref} is a piecewise constant reference, we assume that $i_{ref}(k)$ at time k remains unchanged for the duration of the prediction horizon.

Next, we indicate how the future feedforward input can be obtained, so that constraints on the sum of the MPC control input and the feedforward one can be constrained. The transfer function of the feedforward controller has the following representation:

$$C_{FF}(z) = \frac{U_{FF}(z)}{I_{ref}(z)} = \frac{\sum_{m=0}^{p} b_m z^{-m}}{1+\sum_{n=1}^{q} a_n z^{-n}}. \tag{10}$$

Therefore, the feedforward input signal $u_{FF}(k)$ at time k can be derived using the following difference equation

$$u_{FF}(k) = \sum_{m=0}^{p} b_m i_{ref}(k-m) - \sum_{n=1}^{q} a_n u_{FF}(k-n). \tag{11}$$

The predicted feedforward input signal $\bar{u}_{k+i|k}$ at time $k+i$ is

$$\bar{u}_{i|k} = \sum_{m=0}^{p} b_m i_{ref}(k) - \sum_{n=1}^{q} a_n \bar{u}_{i-n|k}. \tag{12}$$

The varying sequence of the predicted feedforward inputs $\overline{U}_k$ can be expressed as

$$\overline{U}_k = \left[\bar{u}_{0|k}^\top, \dots, \bar{u}_{N-1|k}^\top\right]^\top, \tag{13}$$

where $\bar{u}_{0|k} = u_{FF}(k)$. In this paper, a constant sequence of the predicted feedforward inputs is also considered to compare with (13), which is expressed as

$$\bar{u}_{i|k} = u_{FF}(k), \quad \overline{U}_k = \left[u_{FF}(k)^\top, \dots, u_{FF}(k)^\top\right]^\top. \tag{14}$$

Similarly to the LQR, the MPC cost function is

$$\begin{aligned} J(\hat{x}(k), U_k) &= (x_{N|k} - x_s)^\top P (x_{N|k} - x_s) + \\ &\sum_{i=0}^{N-1} (x_{i|k} - x_s)^\top Q (x_{i|k} - x_s) + (u_{i|k} - u_s)^\top R (u_{i|k} - u_s), \end{aligned} \tag{15}$$

where P is the terminal penalty and Q and R are the stage penalties. Based on the cost function (15), the system model (2) and the sequence of the predicted feedforward inputs (13) or (14), the tracking MPC can be formulated as the following optimization problem:

$$\min_{U_k} J(\hat{x}(k), U_k) \tag{16a}$$

$$\text{s.t. } x_{i+1|k} = A x_{i|k} + B u_{i|k}, \qquad \forall i = 0, \dots, N-1, \tag{16b}$$

$$u_{min} \le u_{i|k} + \bar{u}_{i|k} \le u_{max}, \qquad \forall i = 0, \dots, N-1, \tag{16c}$$

$$x_{0|k} = \hat{x}(k). \tag{16d}$$

It can be observed that the prediction model (16b) does not include the feedforward input, because the feedforward controller and the MPC controller are designed with different control goals in mind. The feedforward controller only inverts the load part of the plant, while the MPC controller ensures the closed-loop performance of the complete plant, and compliance with constraints. The sequence of the predicted feedforward inputs $\overline{U}_k$ is included in the MPC formulation via (16c) and updated at every instant k.

The optimal solution of (16) is denoted as $u_{i|k}^*$, and the total system input at time k is calculated as $u(k) := u_{0|k}^* + \bar{u}_{0|k}$. Hence, the MPC anticipates possible duty-cycle constraints violation due to the feedforward controller and acts to prevent saturation. In the simulation section, we will show that this has a significant impact on position precision.

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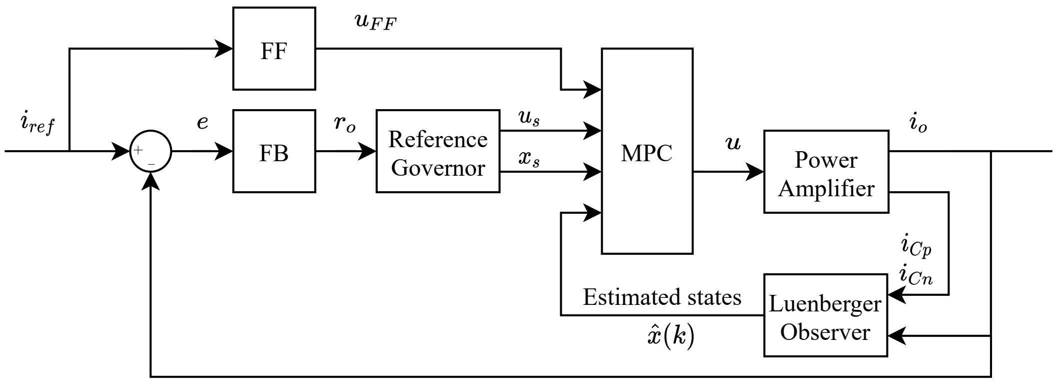

Fig. 4: MPC cascaded current control scheme with $u(k) = u^*_{0|k} + u_{FF}(k)$.

B. FGM solver design

In general, the state prediction for linear MPC can be obtained [16] as:

$$X_k = \Phi\hat{x}(k) + \Gamma U_k, \tag{17}$$

where $X_k = \left[x_{1|k}^\top, \ldots, x_{N|k}^\top\right]^\top$ is the sequence of the predicted states. Substituting (17) into the MPC cost function and removing constant terms yields:

$$\begin{aligned} \min_{U_k} \quad & J(\hat{x}(k), U_k) = \frac{1}{2} U_k^\top G U_k + U_k^\top F(\hat{x}(k)) \\ \text{s.t.} \quad & U_k \in \mathbb{U} := \{U_{min} \le U_k \le U_{max}\}, \end{aligned} \tag{18}$$

where

$$\begin{aligned} G = 2(\Psi + \Gamma^\top \Omega \Gamma),\ & F(\hat{x}(k)) = 2(\Gamma^\top \Omega (\Phi\hat{x}(k) - X_s) - \Psi U_s), \\ X_s = \left[x_s^\top, \ldots, x_s^\top\right]^\top&, U_s = \left[u_s^\top, \ldots, u_s^\top\right]^\top, \\ U_{min} = \Big[(u_{min} - \bar{u}_{0|k})^\top&, \ldots, (u_{min} - \bar{u}_{N-1|k})^\top\Big]^\top, \\ U_{max} = \Big[(u_{max} - \bar{u}_{0|k})^\top&, \ldots, (u_{max} - \bar{u}_{N-1|k})^\top\Big]^\top. \end{aligned} \tag{19}$$

Since only input constraints are considered in the MPC formulation, the fast gradient method is proposed to solve the MPC problem [17]. The FGM solver is summarized in Algorithm 1.

Algorithm 1 FGM for input-constrained MPC [17]

Initialization: $\lambda_{max} = max\{\lambda(G)\}$, $\lambda_{min} = min\{\lambda(G)\}$, $0 < \sqrt{\lambda_{min}/\lambda_{max}} \le \alpha_0 < 1$, ε, $y_0 = z_0$

1: **for** $k = 0 : k_{max}$ **do**
2: $\quad z_{k+1} = [y_k - \lambda_{max}^{-1}(G y_k + F(\hat{x}(k)))]_{\mathbb{U}}$;
3: $\quad$ **if** $\|y_k - z_{k+1}\|^2 \le \varepsilon$ **then**
4: $\quad\quad$ **Break**;
5: $\quad$ **end if**
6: $\quad \alpha_{k+1} = (\sqrt{(\alpha_k^2 - \frac{\lambda_{min}}{\lambda_{max}})^2 + 4\alpha_k^2} - \alpha_k^2 + \frac{\lambda_{min}}{\lambda_{max}})/2$;
7: $\quad \beta = \alpha_k(1 - \alpha_k)/(\alpha_k^2 + \alpha_{k+1})$;
8: $\quad y_{k+1} = z_{k+1} + \beta_k(z_{k+1} - z_k)$;
9: $\quad k \leftarrow k + 1$;
10: **end for**

Usually, z_0 can be initialized either by warm-starting or cold-starting. The shifted sequence warm-starting re-uses the solution of MPC from the previous time instant $k-1$. In this paper, z_0 is initialized with the unconstrained solution $U^*_{uc} = -G^{-1}F(\hat{x}(k))$ from (18), which results in a smaller number of iterations. Because the optimization problem only contains "box" constraints, the projection operation $[v]_{\mathbb{U}} = \arg\min_{z \in \mathbb{U}} \|v - z\|^2$ in step 2 can be simplified as $p = [v]_{\mathbb{U}}$,

$$p_i = \min\{\max\{v_i, U_{min,i}\}, U_{max,i}\}, \forall i = 0, \ldots, M-1, \tag{20}$$

where M denotes the size of U_k. The original FGM algorithm from [17] uses a different stopping criterion in step 3, i.e. $\frac{1}{2}(\frac{1}{\lambda_{min}} - \frac{1}{\lambda_{max}})\|\lambda_{max}(y_k - z_{k+1})\|^2 \le \varepsilon$. In our application, due to the large difference between λ_{min} and λ_{max}, this stopping criterion leads to an unnecessarily large number of iterations. The simpler stopping criterion adopted in this paper results in maximally 5 iterations during all simulations, as detailed in the simulation section.

IV. Simulation Results

A. Motion control system and simulation setup

In this paper, we deal with the current control loop design for a high-precision motion control system, which has a similar outline as shown in Fig. 1. The detailed system structure and the design specifications for the outer position control loop are thoroughly demonstrated in [14]. For the inner current loop, the power amplifier as shown in Fig. 2 is considered and the system parameters are listed in Table I.

TABLE I: Industrial Power Amplifier Parameters

parameter	Symbol	Value
Bus voltage	V_{bus}	360V
Power stage inductance	L	44μH
Power stage capacitance	C	0.4μF
Power stage parasitic resistance	R	62.2μΩ
Load model inductance	L_m	20mH
Load model resistance	R_m	10Ω

Fig. 5 is presented as a benchmark to demonstrate how to evaluate the design performance. For the benchmark, the inner current control scheme is provided from the industrial

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solution, which is optimally designed in the frequency domain. As depicted in Fig. 5a, the outer position controller is optimally designed to track a fixed position reference y_{des}, which belongs to a fourth-order reference profile. Then, the position controller calculates the optimal current reference i_{ref} for the current loop to track. In this paper, the performance of the designed current control loop is evaluated based on the tracking performance of the outer position error, which is defined as $y_{err}(t) = y_{des}(t) - y_{out}(t)$. Fig. 5b presents the corresponding trajectory of y_{err} when the inner loop is controlled based on the benchmark scheme. It is observed that when the current reference i_{ref} is varying, the position error y_{err} increases due to the duty-cycle saturation issue. In addition, an exposure window is highlighted in Fig. 5b. In lithography, the position tracking performance is important when the wafer is moving at a constant speed. Fig. 5c illustrates the evaluated position tracking performance based on the following criteria [3]:

$$\int_{t-T/2}^{t+T/2} w_{th}(\tau)d\tau = 1, \tag{21a}$$

$$\mathrm{MA}(t) = w_{th}(t) * y_{err}(t), \tag{21b}$$

$$\mathrm{MSD}(t) = \sqrt{w_{th}(t) * y_{err}^2(t) - \mathrm{MA}^2(t)}, \tag{21c}$$

$$\mathrm{MRMS}(t) = \sqrt{\mathrm{MA}(t)^2 + \mathrm{MSD}(t)^2}, \tag{21d}$$

where $w_{th}(t)$ is a filter window that is limited by the slit time T. The peak values of $|\mathrm{MA}|$, MSD and MRMS during the exposure window (22) are regarded as the benchmark and the cascaded current control schemes are proposed to improve this performance.

$$|\mathrm{MA}|_\mathrm{p} = 2.77\mathrm{pm},\ \mathrm{MSD_p} = 32.34\mathrm{pm},\ \mathrm{MRMS_p} = 32.45\mathrm{pm} \tag{22}$$

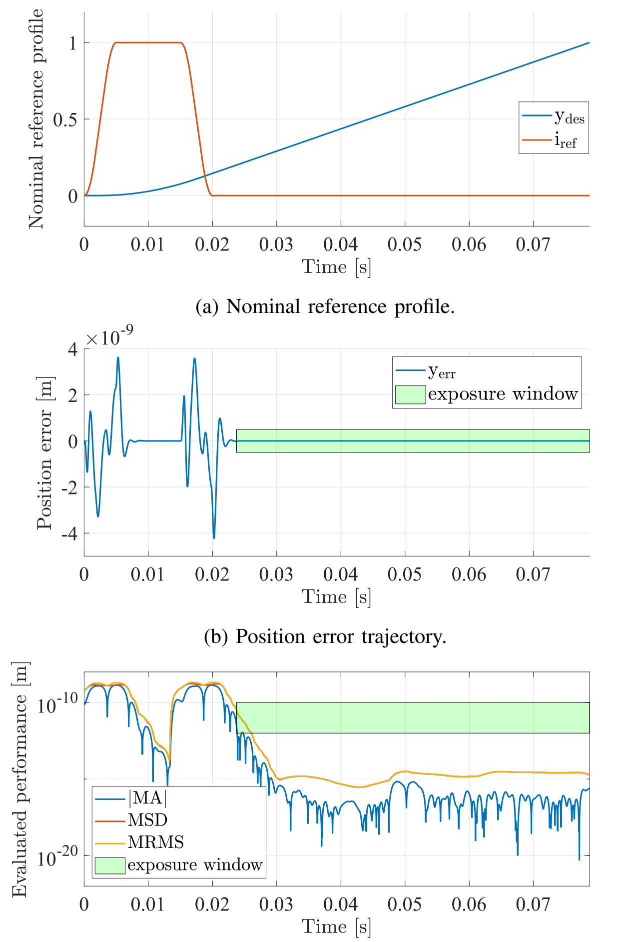
All the simulations of the high-precision motion control system are performed in MATLAB/Simulink R2022b with the following specifications: Desktop computer, AMD Ryzen 7 3800XT CPU 4.0GHz, 32GB RAM, 64–bit OS. Different performances of the cascaded current control schemes are compared when the feedback and feedforward controllers in the current control loop as well as the outer position control loop maintain the same.

TABLE II: MPC Setup

parameter	Symbol	Value
Sampling frequency	f_s	400kHz
Prediction horizon	N	10
Weighting matrix	Q	$\mathrm{diag}(q\frac{L}{L_m}, q\frac{C}{L_m}, q\frac{L}{L_m}, q\frac{C}{L_m}, 1)$
Weighting matrix	R	$\mathrm{diag}(10, 10)$
Weighting matrix	P	P_{LQR} or 10^3Q
Input constraints	$\mathbb{U}$	$u_{i\|k} + \bar{u}_{i\|k} \in [0,1]^2$

B. MPC controller performance analysis

In this subsection, the performance of the MPC cascaded current control scheme as shown in Fig. 4 is analyzed. We first compare the MPC performances with different predicted feedforward input sequences. Next, the effect of the different weighting values on the position tracking performance is demonstrated. Finally, we validate the performance of the FGM solver. The MPC setup is listed in Table II. The position loop is sampled with 25µs, which is 10 times larger compared to T_s. Thus, the prediction horizon N is set to 10. Fig. 6 compares the position error trajectories of the MPC cascaded current control scheme when the predicted feedforward sequences are different. note that the matrix Q depends on a scalar weighing parameter q, which is set as $q = 10^4$ and the terminal weighting matrix is $P = 10^3Q$. As discussed in Section III, two predicted feedforward sequences are proposed: the constant one (14) and the varying one (13). It is noticed that the MPC formulation with varying predicted feedforward sequence would result in a relatively smaller position error before the exposure window, owing to the fact that the predicted future information for this case is more accurate. Since the evaluated tracking performances



(a) Nominal reference profile.

(b) Position error trajectory.

(c) Evaluated performance.

Fig. 5: Evaluation of the benchmark current control scheme.

TABLE III: Position error: different predicted FF sequences

q	$\|\mathrm{MA}\|_\mathrm{p}$ [pm]	$\mathrm{MSD_p}$ [pm]	$\mathrm{MRMS_p}$ [pm]
Constant FF	2.04	11.76	11.89
Varying FF	2.24	12.60	12.80

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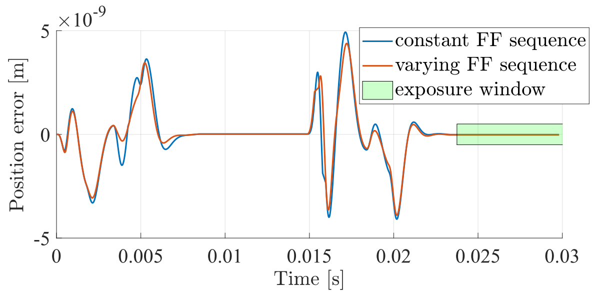


Fig. 6: Position error: different predicted feedforward inputs.

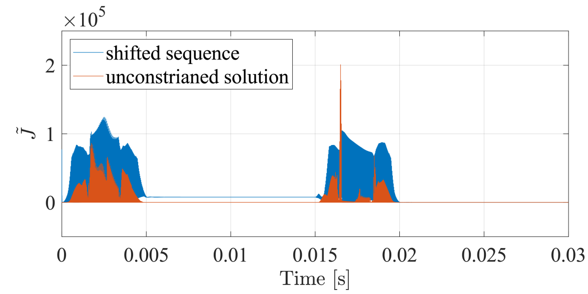


Fig. 8: FGM optimality with different initialization.

shown in Table III for both cases are very close, for all the following MPC setups, the predicted feedforward input sequence is constructed based on the varying case (13).

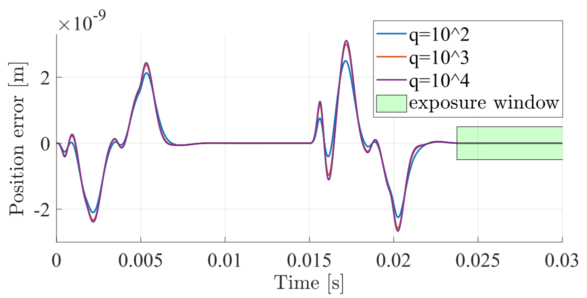


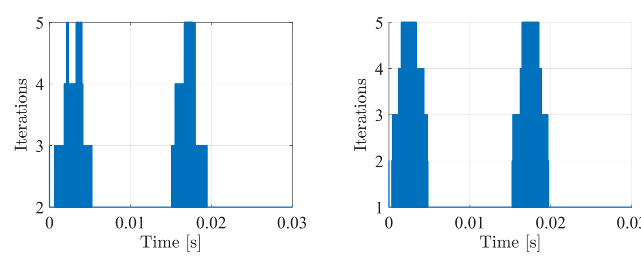
Fig. 7: Position error: different matrices Q.

Fig. 7 compares the position error trajectories when the weighting matrix Q is assigned with different q. The terminal weighting matrix is selected as $P = P_{LQR}$, which is derived by solving the algebraic Riccati equation of the LQR. As q increases, i.e., less penalty on the output current i_o, the position error becomes larger before the exposure window. However, as demonstrated in Table IV, a higher q results in smaller evaluated $|\text{MA}|_{\text{peak}}$, MSD_{peak} and $\text{MRMS}_{\text{peak}}$. This demonstrates that embedding the full system information in controller design can contribute to performance improvement. For the proposed FGM algorithm, the performance is

TABLE IV: Position error: different matrices Q

q	$\lvert\text{MA}\rvert_{\text{p}}$[pm]	MSD_{p}[pm]	MRMS_{p}[pm]
10^2	13.26	17.55	22.00
10^3	10.69	15.04	18.45
10^4	10.10	14.50	17.66

compared for different initialization sequences with respect to z_0. The MPC optimal solution can be computed by solving a quadratic program (QP) online, in simulation, where timing constraints may be violated. The deviation from the optimal MPC-QP solution of the proposed FGM algorithm is evaluated based on the difference of the cost value (15) as demonstrated in Fig. 8. The cost value difference is defined as $\tilde{J}(k) = J(k) - J^*(k)$, where $J^*(k)$ represents the optimal cost for MPC-QP. Then, utilizing the corresponding open-loop state trajectory, the cost value $J(k)$ is calculated by solving the MPC using the FGM algorithm. In general, a smaller $\tilde{J}(k)$ is achieved when FGM is initialized with the unconstrained solution sequence. Fig. 9 compares the number of iterations, while the averaged numbers of 2.11 and 1.10 are obtained for the shifted sequence case and the unconstrained solution case respectively. It indicates that the initialization with the unconstrained solution improves the convergence speed as well, besides optimality. The mean computation time of the MPC problem with such initialization in the Simulink environment is approximately 65µs. However, the implementation of such an algorithm in C and FPGA would considerably increase the computation speed, especially since this algorithm only involves basic operations. In [9], a similar MPC problem with a long prediction horizon was solved on FPGA in the range of a few µs. Hence it is expected to meet the 2.5µs target for the considered amplifier application in this paper. Future work will deal with optimized FPGA implementation of the developed MPC controller.



(a) Shifted sequence

(b) Unconstrained solution

Fig. 9: FGM number of iterations with different initialization.

Next, we compare LQR and MPC based on the evaluated performances of the position tracking. We first demonstrate that MPC with FF preview can handle saturation issues. Then we present the optimally tuned MPC with the modified terminal penalty which gives the best tracking performance. Fig. 10 compares the position error trajectories of the LQR and the MPC when the same weighting matrices Q ($q = 10$) and R are assigned. The terminal matrix P of the MPC is selected as P_{LQR}, such that the MPC would behave the same as the LQR if the control input is not saturated. It is intuitive that the LQR generates a large position error before the exposure window, while the MPC keeps a similar position error level as depicted in Fig. 7. When the current reference i_{ref} is varying, the feedforward controller would react to the reference change and contribute to a saturated input.

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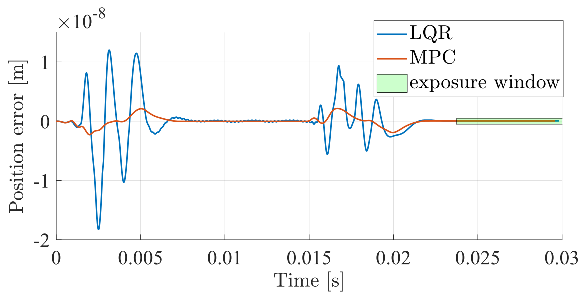

Fig. 10: Position error: LQR versus MPC.

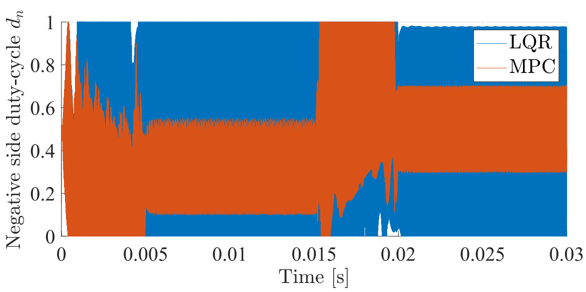

Fig. 11: Total input (d_n): LQR versus MPC with FF preview.

Fig 11 presents the compared total inputs on the negative side (d_n), the LQR loses its control performance due to saturation. Since MPC ensures that the total control input meets constraints, it leads to less position error and better tracking (as shown in Table V). Apart from the fact that MPC

TABLE V: Position tracking error: LQR versus MPC

q	$\lvert\mathrm{MA}\rvert_\mathrm{p}$ [pm]	MSD_p [pm]	MRMS_p [pm]
LQR	17.98	35.64	39.92
MPC	15.68	20.07	25.46

can eliminate the saturation problem, we can also assign a different P to further improve the tracking performance. Table VI presents the evaluated position tracking performances with different current control schemes (Benchmark, LQR cascaded scheme, MPC cascaded scheme). For each control scheme, the control setup is optimally tuned. For both LQR and MPC, Q is selected with $q = 10^4$. The terminal weighting matrix P for MPC is $P = 10^3 Q$. A significant improvement in the position tracking performance can be achieved by the MPC cascaded control scheme.

TABLE VI: Overall position tracking error comparison

	$\lvert\mathrm{MA}\rvert_\mathrm{p}$ [pm]	MSD_p [pm]	MRMS_p [pm]
Benchmark	2.77	32.34	32.45
LQR	10.10	14.50	17.66
MPC	2.24	12.60	12.80

V. Conclusions

In this paper, we considered the current control problem for power amplifiers employed in high-precision positioning systems within lithography machines. The current industrial control solutions and an earlier LQR-based solution have limited performance due to the saturation of the duty-cycle control inputs. We show that by employing a cascaded MPC control structure that anticipates future feedforward control actions, saturation can be prevented. This further results in a significant improvement of the positioning accuracy (at least 2 times). To achieve real-time implementation of the MPC controller, an adapted fast gradient method from the literature was used. Future work will deal with real-life implementation using an FPGA and a prototype power amplifier circuit that is currently under development. In future work, we will consider the design of a data-driven predictive controller, which does not require a state observer. The parameter-varying issue of the load will be investigated as well.

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